

10-Mark Questions & Answers MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Explain the internal architecture of 8086 in detail (BIU, EU, registers, flag register) with a neat labeled diagram.

1. Bus Interface Unit (BIU)

2. Execution Unit (EU)

Q2. Explain the functional pin diagram of 8086 microprocessor in detail, describing the function of each pin

group (address/data bus, control signals, power supply).

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1. **Multiplexed Address/Data Bus (AD0 - AD15):** Time-multiplexed lines carrying lower 16-bit memory address during `T1` and 16-bit data during `T2-T4`.

2. **Ac**

Q3. Explain memory segmentation and banking in 8086 in detail with physical address calculation examples. Also explain why segmentation is used.

1. Why Memory Segmentation is Used

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2. Physical Address Calculation Formula & Example

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- **Example 1 (Instruction Fetch):** `CS = 3000H`, `IP = 0100H`

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3. Memory Banking System (Even & Odd Banks)

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Q4. Explain the functioning of 8086 in both minimum mode and maximum mode with neat diagrams, and state the key differences.

1. Minimum Mode Block Diagram (MN/MX' = 1)

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2. Maximum Mode Block Diagram (MN/MX' = 0)

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Q5. Draw and explain in detail the timing diagrams for read and write operations in both minimum and maximum mode of 8086.

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MINIM

CLK : _ _ _ _ _

| T1

Q6. Explain the interrupt structure of 8086 in detail, including types of interrupts (hardware/software), interrupt vector table, and interrupt servicing procedure.

INTER

Q7. Explain the process of demultiplexing the address/data bus in 8086 with a circuit diagram (using latch 74LS373).

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